

Benchmarking in AI/LLM Research

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Previous benchmarking work :

1. EthosAgentAI: An Ethical, Multi-Model Agentic AI Ecosystem for Fair Hiring and Intelligent Skill Development, aligned with SDG 8 and SDG 9.

2. (Naive-RAG-and-StreamRAG-Benchmarking) <https://github.com/DIP-RO/Naive-RAG-and-StreamRAG-Benchmarking-Application>

TL;DR

- **Strongest, best-evidenced, low/medium-compute opportunity:** a Bangla RAG faithfulness/hallucination benchmark. A dedicated verification (Aug 2026) found no published Bangla/Bengali resource combining RAG + hallucination/faithfulness: BenHalluEval (non-RAG), LegalRAG (retrieval-quality only), and BanglaMedQA (accuracy-only) all exist separately, and the multilingual RAG-faithfulness benchmark [MEMERAG \(ACL 2025\)](#) explicitly excludes Bengali while a proven template already exists in Turkish (Turk-LettuceDetect) and English (RAGTruth).
- **English-language benchmarking is largely saturated or crowded** (HumanEval, MMLU, GSM8K, RAGTruth, RULER, AgentBench). The defensible, still-open gaps are in low-resource languages, faithfulness/robustness under retrieval, and reproducibility/LLM-as-judge reliability not in yet another English capability leaderboard.
- **Recommended direction:** "BanglaRAGTruth" a Bangla RAG faithfulness benchmark plus a lightweight token-level hallucination detector. It is feasible on API models plus a single mid-tier GPU (the Turkish analogue trained each encoder in ~2 hours on one A100), has a validated methodological template, and targets a genuinely open niche with clear publication potential.

Key Findings

1. **The Bangla RAG + hallucination + faithfulness niche is open.** No published Bangla/Bengali benchmark or dataset combines RAG with hallucination/faithfulness evaluation as of August 2026. The closest multilingual RAG-faithfulness benchmark, [MEMERAG \(Amazon Science, ACL 2025, arXiv:2502.17163\)](#), is built on MIRACL with 2,322 sentence-level faithfulness/relevance annotations across exactly English, German, Spanish, French, and Hindi Bengali is excluded.
2. **Adjacent Bangla work exists but does not fill the niche.** [BenHalluEval \(2026\)](#) covers Bangla hallucination but is explicitly non-RAG; [LegalRAG \(2025\)](#) is Bangla legal RAG but scores retrieval only; [BanglaMedQA/BanglaMMedBench \(2025\)](#) evaluate RAG strategies by accuracy, not faithfulness; [BanglaMATH \(2025\)](#) covers math reasoning.
3. **The methodological template is proven and cheap.** [Turk-LettuceDetect \(Turkish, 2025\)](#) machine-translated RAGTruth and fine-tuned encoder detectors, reaching F1 0.7266, with each model trained in ~2 hours on a single NVIDIA A100 directly transferable to Bangla.
4. **RAG retrieval-strategy findings are genuinely contradictory across papers** (BM25 vs dense), driven by domain a real research opportunity, but crowded in English.

5. LLM-as-judge reliability and multilingual judge transfer are hot and underexplored, especially for low-resource languages where judge reliability collapses.

Details

Area G — Multilingual / Bangla (the focal opportunity)

Established Bangla resources:

- **TituLLMs** — [Nahin et al., Findings of ACL 2025; arXiv:2502.11187](#) first large pretrained Bangla LLMs (1B, 3B); pretraining dataset of approximately 37 billion tokens; extended Llama-3.2 tokenizer; the authors developed five benchmarking datasets to address the lack of Bangla evaluation data.
- **TigerLLM** — [Raihan & Zampieri, ACL 2025 Short Papers; arXiv:2503.10995](#) Bangla LLM family; claims to surpass open alternatives and GPT-3.5 on standard benchmarks (1B and 9B).
- **BnMMLU** — [arXiv:2505.18951](#) massive multitask understanding in Bengali.
- **BEnQA** — (Shafayat et al., 2024) bilingual Bangla science QA/reasoning.
- **BanglaMATH** — [Prama, Danforth, Dodds, MathNLP 2025; arXiv:2510.12836](#) 1.7k Bangla math word problems (grades 6–8); Gemini 2.5 Flash and DeepSeek V3 are the only models to achieve strong performance, with $\geq 80\%$ accuracy across three elementary school grades; both fail robustness under distractors and show language bias vs English (Bangla is spoken by nearly 250 million people). Dataset on GitHub.
- **GSM-Plus-BN** — (Paul et al.) perturbation-based Bangla math robustness; GPT-OSS-20B reached 96.08%.
- **MathShikkha** — [arXiv:2608.08503](#) controlled Bangla CoT study; in-domain CoT gains vanish for strong backbones but CoT protects out-of-domain generalization (+20.1–28.1 points on contamination-audited BanglaMATH).
- **BenHalluEval** — [Adib et al.; arXiv:2605.31483](#) first Bangla hallucination framework; 4 tasks (GQA, code-mixed QA, summarization, reasoning); 12,000 hallucinated candidates; dual-track BenHalluScore ranging 7.72%–55.42%. Explicitly NON-RAG ("no benchmark measures hallucination in LLMs on both native Bengali and Bangla-English code-mixed input").
- **LegalRAG** — [Kabir et al.; arXiv:2504.16121; IEEE](#) bilingual (Bangla/English) legal RAG on Bangladesh Police Gazettes; evaluates retrieval quality, not faithfulness.
- **BanglaRAG** — (Sanju-Basak, GitHub) open-domain Bengali RAG pipeline (fine-tuned ColBERT + BN-RAG-LLaMA3-8b; $\sim 8\%$ enhancement); states there are no benchmarks for Bengali information retrieval.
- **BanglaMedQA / BanglaMMedBench** [arXiv:2511.04560](#) Bangla biomedical QA; evaluate RAG strategies (traditional, zero-shot fallback, agentic, iterative feedback, aggregate) by accuracy.
- **Bengali dialectal bias / RLAIIF** [arXiv:2603.21359](#); Bengali legal LLM-as-judge (Aftahee et al.; [arXiv:2511.05627](#)) LLM-as-judge used for Bangla, but not for RAG faithfulness.

- **BanglaGuard** — [referenced in arXiv:2608.02941](#) early dedicated Bengali safety pipeline.

Comparison to other low-resource languages: Turkish has [Turk-LettuceDetect \(arXiv:2509.17671\)](#) (RAG hallucination detection; ModernBERT F1 0.7266, trained ~2 hrs/model on one A100). Persian has [PerHalluEval \(arXiv:2509.21104\)](#).

Correction to the assignment framing: PerHalluEval is a QA+summarization hallucination benchmark and is NOT RAG-specific; therefore the low-resource RAG-hallucination analogue exists mainly in Turkish, which strengthens the Bangla opportunity rather than weakening it.

Area A — General LLM benchmarking

Saturation is pervasive: MMLU saturated above 90% and contamination is well-documented (rephrased-sample leakage). Standard suites: MMLU, MMLU-Pro, GPQA, GSM8K, BBH. Uncontrolled variables include prompt formatting, few-shot count, decoding settings, and silent model-version drift.

Area B — RAG benchmarking

- **RAGTruth** [Niu et al., ACL 2024; arXiv:2401.00396](#) 17,790 responses to 2,965 prompts from 6 LLMs (GPT-3.5-Turbo-0613, GPT-4-0613, Llama-2-7B/13B/70B-Chat, Mistral-7B-Instruct) across QA, data-to-text, and summarization; word- and case-level hallucination annotations; official split 15,090 train / 2,700 test; fine-tuned Llama-2-13B competitive on span detection.
- **RAGBench** [arXiv:2407.11005](#) 100k examples, 5 industry domains, explainable (TRACe metrics).
- **RAGAS** [Es et al., EACL 2024; arXiv:2309.15217](#) reference-free; defines Faithfulness, Answer Relevance, Context Relevance (library adds Context Precision/Recall). Validated in English on WikiEval; language-agnostic in principle but no native Bangla validation.
- **ARES** [Saad-Falcon et al., NAACL 2024; arXiv:2311.09476](#) context relevance, answer faithfulness, answer relevance; fine-tuned LM judges + prediction-powered inference; its judge is 59.9 percentage points higher than RAGAS in context relevance and 14.4 points higher in answer relevance.
- **Bi'an** [arXiv:2502.19209](#) bilingual (English/Chinese) RAG hallucination benchmark + detector.

Retrieval-strategy contradictions (genuine):

- [T2-RAGBench financial study \(arXiv:2604.01733\)](#) BM25 beats dense (+26.7% Recall@5 gap), hybrid+cross-encoder rerank dominates (Recall@5 0.816).
- [Biomedical study \(arXiv:2605.02520\)](#) cross-encoder rerank best (composite 0.827) but dense a strong baseline (0.822), multi-query harmful; faithfulness uniformly 0.893–0.902 and near-perfect (0.978) under no-context a measurement artifact showing faithfulness must be read with answer relevance.

- [Nutrition study \(arXiv:2604.25928\)](#) dense (78.2%) > BM25 (72.1%) > hybrid (71.9%).
[CRUD-RAG \(arXiv:2401.17043\)](#) (Chinese) task-dependent.

Cause: domain lexical density and query type.

Area C LLM agents

AgentBench (Liu et al., 2023), WebArena, SWE-bench, GAIA, ToolBench, τ -bench, AgentBoard, AgentGym. Metrics: success rate, task completion, tool-use accuracy, reward. Survey ([arXiv:2507.21504](#)) systematizes tool use, planning, memory, multi-agent. Agent Planning Benchmark ([arXiv:2606.04874](#)) isolates planning vs execution. High compute and low reproducibility (scaffold-dependent) make this a poor student fit.

Area D Reasoning

GSM8K, MATH, GPQA; reasoning models (o1-style, DeepSeek-R1); CoT improves math/symbolic selectively (e.g., +12.3% math, +14.2% symbolic in one study; minimal on commonsense). Contamination motivates live benchmarks (LiveCodeBench).

Area E Hallucination

RAGTruth, HaluEval, HaluBench, FaithBench, AggreFact, TofuEval, FACTS Grounding; Vectara HHEM + FaithJudge leaderboard (Tamber et al., EMNLP 2025 Industry; [arXiv:2505.04847](#)) LLM-as-judge with human-annotated few-shot examples improves RAG faithfulness evaluation; LettuceDetect ([arXiv:2502.17125](#), token-level, 30–60 examples/sec on one A100, near-SOTA at a fraction of cost).

Area F LLM-as-a-judge

Documented biases: position, verbosity, self-enhancement/self-preference, family, cross-lingual degradation. A large-scale study ([arXiv:2606.19544](#)) reports "kappa deflation" (raw agreement overstates chance-corrected discrimination by 33–41pp), non-transferable rankings (shifts up to 14 positions), and much-reduced verbosity bias (<0.011) vs the 20–40% reported in 2023. Cross-lingual judge reliability collapses in low-resource languages (BabelJudge, [arXiv:2606.22329](#); Cross-Lingual LLM-Judge Transfer, [arXiv:2603.18557](#)).

Area H Safety

DecodingTrust, HarmBench, JailbreakBench, SafetyBench, StrongREJECT, RealToxicityPrompts, PolygloToxicityPrompts, R-Judge (agent safety). Multilingual safety is English-centric; low-resource prompting acts as a safety stressor (e.g., Batak Toba correctness dropping from 0.68 to 0.06). Bangla: BanglaGuard, and derogatory-speech comprehension–containment decoupling ([arXiv:2608.02941](#)).

Area I Code

HumanEval and MBPP saturated (>90%) and contaminated (ID-based memorization on SWE-bench Verified); SWE-bench Verified is the practical repo-level standard; LiveCodeBench for contamination-free live eval; mHumanEval-Bangla and MBPP-Bangla/TigerCoder extend to Bangla.

Area J Long-context

Needle-in-a-Haystack, RULER (NVIDIA, 13 tasks/4 categories), LongEval, U-NIAH (unified RAG vs LLM NIAH, ACM TOIS; RAG wins 82.58% over direct answering; smaller LLMs benefit more from RAG). Effective context length << advertised; lost-in-the-middle persists; multi-needle scores 15–40 points below single-needle.

Area K Multimodal

MMBench, SEED-Bench, MMStar, MMMU; multimodal RAG (MRAG-Bench, VisR-Bench, FATHOMS-RAG, Visual-RAG, M2RAG [arXiv:2502.17297](https://arxiv.org/abs/2502.17297)); multimodal hallucination detection; visual retrievers outperform textual; models struggle with non-textual elements and fine-grained grounding.

Area L Research agents

AstaBench (Ai2), DeepResearch Bench (100 PhD-level tasks), ScholarQABench (2,967 queries), ResearcherBench (65 questions), LitSearch, ReportBench, ResearchArena, AutoResearchBench (SOTA only ~9.39% Deep Research accuracy). Report evaluation is intrinsically hard (no single gold answer).

Metrics comparison (condensed)

- **Accuracy / Exact Match** : clear for closed-form MCQ/math; misleading for open generation and cross-lingual output.
- **F1 / token-F1** : best for span-level hallucination detection; threshold-sensitive.
- **ROUGE / BLEU** : cheap, weak semantic signal; misleading for factuality.
- **Recall@K / MRR / nDCG** : retrieval quality only; blind to generation faithfulness.
- **Faithfulness / Groundedness** : core RAG hallucination signal; unreliable read alone (defaults near-perfect with no context).
- **Answer Relevance** : must be paired with faithfulness.
- **Hallucination Rate** : intuitive but definition-dependent (factual vs contextual).
- **Pass@K** : code correctness; contamination-sensitive.
- **Success / Task Completion / Tool-Use Accuracy** : agents; scaffold-dependent, low reproducibility.
- **Safety Rate / Robustness** : critical but protocol-fragmented.
- **Latency / Token Usage / Cost** : rarely reported; a real, easy-to-address gap.
- **Human Preference** : gold but expensive; the LLM-as-judge proxy carries measurable bias.

Methodological problems (evidence-based)

Saturation (MMLU, HumanEval); contamination/leakage (SWE-bench ID memorization; HumanEval rephrasing inflates scores 15–20pp); poor reproducibility (prompts, model versions, retrieval configs, hyperparameters, hardware); cherry-picked datasets/metrics; lack of statistical significance testing; English-centric evaluation; LLM-judge bias; human-vs-LLM disagreement (kappa deflation); missing cost/latency/robustness/adversarial reporting.

Genuine contradictions (as research opportunities)

- **BM25 vs dense retrieval:** domain-dependent reversals (financial/legal/code/SE favor BM25; paraphrase-heavy general QA favors dense). Likely cause: lexical constraint density.
- **CoT value:** BanglaMATH/MathShikkha show in-domain CoT gains can vanish for strong backbones but protect out-of-domain generalization — depends on backbone strength and contamination auditing.
- **Verbosity bias in judges:** 20–40% (2023) vs <1.1% (2026) — model-generation-dependent, cautioning against citing old bias magnitudes.

Recommendations

Decision matrix

Scored 1–5, higher is better; Overall = weighted mean favoring novelty, data availability, and compute feasibility.

Area	Novelty	Lit. Avail.	Data Avail.	Impl. Ease	Low Compute	Eval Ease	Pub. Potential	Overall
Bangla RAG faithfulness/hallucination	5	4	4	4	5	3	5	4.4
Bangla LLM-as-judge reliability	4	4	4	4	5	3	4	4.1
LLM-as-judge bias (English)	3	5	5	4	5	3	3	3.9

Area	Novelty	Lit. Avail.	Data Avail.	Impl. Ease	Low Compute	Eval Ease	Pub. Potential	Overall
RAG retrieval-strategy (English)	2	5	5	4	4	4	3	3.6
Long-context vs RAG	3	4	4	3	3	3	4	3.4
Hallucination (English)	2	5	5	3	4	3	3	3.4
Code LLM	2	5	5	3	3	4	2	3.3
Agents / research agents	4	4	3	2	2	2	4	3.0
Multimodal RAG	4	3	3	2	2	2	4	2.9

Thresholds that would change the recommendation: if a Bangla RAG-faithfulness benchmark is published before you start (monitor arXiv/ACL BLP workshop), pivot to (a) extending it to code-mixed/dialectal Bangla, or (b) Bangla LLM-as-judge reliability, which remains open.

Compute classification

- **Low:** Bangla RAG faithfulness benchmark (API judges + one GPU for the encoder detector); Bangla LLM-as-judge reliability; RAG retrieval-strategy/metric reproducibility audit.
- **Medium:** fine-tuning small Bangla detectors at scale; long-context vs RAG for Bangla long docs.
- **High:** agent training, multimodal training — avoid for a student without clusters.

Dataset table (selected, with student-suitability)

Dataset	Area	Size	Task	Lang	Metrics	Open	Note for student
RAGTruth	RAG hallucination	17,790 resp.	QA/D2T/Summary	EN	span F1	Yes	Base to — translate/adapt ideal
Turk-Lettuce Detect data	RAG hallucination	17,790 (MT)	QA/D2T/Summary	TR	token F1, AUC, ROC	Yes	Direct template
MEMERAG	RAG faithfulness	2,322 sent.	QA	EN/DE/ES/FR/HI	faith/reliability	Yes	Extend to Bengali

Dataset	Area	Size	Task	Lang	Metrics	Open	Note for student
BenHalluEval	Hallucination (non-RAG)	12,000 cand.	GQA/CM-QA/Summ/Reas	BN	BenHalluScore	Yes	Seed source for Bangla
LegalRAG	Bangla RAG (retrieval)	—	Legal QA	BN/EN	retrieval	Yes	Retrieval-only baseline
BanglaMATH	Reasoning	1.7k	Math WP	BN	accuracy	Yes	Robustness template
RULER	Long-context	synthetic	13 tasks	EN	accuracy	Yes	For long-context vs RAG
SWE-bench Verified	Code	500	Repo issue	EN(cod e)	resolve%	Yes	High compute

Top-5 experimental designs (condensed)

- **1. BanglaRAGTruth (recommended).** RQ: Can a Bangla RAG faithfulness benchmark + lightweight detector reliably flag unsupported spans? Method: translate/adapt RAGTruth to Bangla + construct native Bangla RAG QA from BanglaWikipedia/news + BenHalluEval seeds; human-verify faithfulness spans; fine-tune BanglaBERT/ModernBERT/EuroBERT detectors; compare vs RAGAS faithfulness and prompt-based LLM judge. Metrics: token/response F1, AUROC, balanced accuracy, cost/latency. Stats: paired bootstrap, McNemar.
- **2. Bangla LLM-as-judge reliability** position/verbosity/self-preference bias + human agreement (Cohen's κ) for Bangla judging.
- **3. RAG faithfulness-metric reproducibility audit** (any language) controlled RAGAS vs LLM-judge vs encoder on existing English RAGTruth, with significance testing and cost reporting.
- **4. Long-context vs RAG for Bangla long-doc QA** RULER-style adaptation.
- **5. Bangla safety/jailbreak robustness benchmark** extend HarmBench/JailbreakBench prompts.

Final five recommendations

- **Topic 1 (Strongest Overall):** BanglaRAGTruth Bangla RAG faithfulness benchmark + detector.
- **Topic 2 (Most Novel):** Bangla RAG hallucination detection under Bangla-English code-mixing (no code-mixed RAG faithfulness resource exists anywhere).
- **Topic 3 (Easiest to Implement):** Reproducibility audit of RAG faithfulness metrics (RAGAS vs LLM-judge vs encoder) on existing English RAGTruth ,no new data needed.
- **Topic 4 (Best Publication Potential):** Multilingual RAG faithfulness extending MEMERAG to Bengali (+ Hindi/Urdu cross-lingual) venue fit at ACL/EMNLP multilingual tracks.

- **Topic 5 (Best Bangla/Low-Resource):** BanglaRAGTruth (same as Topic 1).

My Recommended Research Direction

Proposed Title: BanglaRAGTruth A Faithfulness and Hallucination Benchmark and Lightweight Detector for Bengali Retrieval-Augmented Generation.

Research Gap (Established → Evidence → Contribution)

Established RAG hallucination benchmarks (RAGTruth), detectors (LettuceDetect), and low-resource adaptations (Turk-LettuceDetect) exist; Bangla has hallucination (BenHalluEval), legal RAG (LegalRAG), and biomedical RAG (BanglaMedQA) resources.

Evidence-based gap verified Aug 2026: no Bangla resource combines RAG + faithfulness/hallucination; MEMERAG excludes Bengali; BenHalluEval is non-RAG; LegalRAG scores retrieval only.

Proposed contribution the first Bangla RAG faithfulness benchmark, an open token-level detector, and the first code-mixed RAG faithfulness evaluation.

Problem Statement

Bengali RAG systems can produce unsupported or contradictory claims even when given relevant retrieved context, yet no resource measures or detects this.

Research Questions

- RQ1 How often, and in what patterns, do LLMs hallucinate in Bengali RAG across QA, summarization, and data-to-text?
- RQ2 Can a fine-tuned encoder detector match/beat prompt-based LLM judges and RAGAS faithfulness in Bengali at lower cost?
- RQ3 Does Bangla-English code-mixing increase hallucination and degrade detection?

Hypotheses

- H1 hallucination rates are higher in Bangla than English for the same models.
- H2 a fine-tuned encoder detector is $\approx$ or $>$ LLM-judge at far lower cost/latency.
- H3 code-mixing increases hallucination and lowers detector F1.

Dataset

- (a) Machine-translate RAGTruth to Bangla (as Turk-LettuceDetect did for Turkish).
- (b) Construct native Bangla RAG QA from BanglaWikipedia and news corpora.
- (c) Draw code-mixed items from BenHalluEval seeds; human-verify faithfulness spans on a stratified sample.

Models

Generators GPT-class, Gemini 2.5 Flash, DeepSeek V3, TituLLM/TigerLLM.

Detectors BanglaBERT, multilingual ModernBERT/EuroBERT.

Baselines

RAGAS faithfulness, prompt-based GPT/Gemini LLM-judge, multilingual LettuceDetect (zero-shot transfer).

Experimental Pipeline

generation under fixed retrieval configs → span annotation → detector fine-tuning → head-to-head comparison vs baselines → ablations → error analysis.

Metrics

token-level and response-level F1, AUROC, balanced accuracy; plus cost (USD/1k) and latency (examples/sec).

Ablation Study

encoder architecture; context length (up to 8,192 tokens); native vs code-mixed; machine-translated vs native-authored data; effect of judge model choice.

Error Analysis

by hallucination type (evident/subtle conflict, baseless introduction), by task, and by entity rarity.

Expected Results

encoder F1 ~0.65–0.75 (by Turkish analogy, where ModernBERT reached 0.7266); LLM judges accurate but costly; code-mixing measurably degrades faithfulness a novel finding.

Expected Contribution

first Bangla RAG faithfulness benchmark + open detector + released dataset; first code-mixed RAG faithfulness evaluation; a reusable low-compute recipe for other low-resource languages.

Limitations

machine-translation artifacts; annotation cost/subjectivity; potential judge circularity (mitigate with cross-model judges, as in recent Bengali VLM work).

8–12 Papers to Read First

- RAGTruth [arXiv:2401.00396](https://arxiv.org/abs/2401.00396)
- Turk-LettuceDetect [arXiv:2509.17671](https://arxiv.org/abs/2509.17671)
- RAGAS [arXiv:2309.15217](https://arxiv.org/abs/2309.15217)
- ARES [arXiv:2311.09476](https://arxiv.org/abs/2311.09476)
- MEMERAG [arXiv:2502.17163](https://arxiv.org/abs/2502.17163)
- BenHalluEval [arXiv:2605.31483](https://arxiv.org/abs/2605.31483)
- LegalRAG [arXiv:2504.16121](https://arxiv.org/abs/2504.16121)

- FaithJudge / Vectara [arXiv:2505.04847](https://arxiv.org/abs/2505.04847)
- Bi'an [arXiv:2502.19209](https://arxiv.org/abs/2502.19209)
- PerHalluEval [arXiv:2509.21104](https://arxiv.org/abs/2509.21104)
- BanglaMATH [arXiv:2510.12836](https://arxiv.org/abs/2510.12836)
- TituLLMs [arXiv:2502.11187](https://arxiv.org/abs/2502.11187)

Caveats

- **Negative claims cannot be proven absolutely.** "No Bangla RAG-faithfulness benchmark exists" reflects thorough arXiv/ACL/BLP-workshop searching as of August 2026; monitor for new preprints before submission and pivot per the thresholds above if one appears.
- **Correction to the assignment premise:** the Persian "RAG hallucination" comparator (PerHalluEval) is actually non-RAG; the true RAG-specific low-resource analogue is Turkish (Turk-LettuceDetect). This strengthens, not weakens, the Bangla case.
- **Recency risk:** several cited 2026 arXiv IDs are very recent preprints, not yet peer-reviewed treat as provisional and re-verify in primary PDFs.
- **Metric provenance:** a few figures were drawn from secondary summaries and were cross-checked; where a metric was absent from a paper it is marked "Not reported." Exact RAGTruth counts vary slightly across citing papers (17,790 responses / 2,965 prompts is the primary-source figure).